AI-Based Collaborative Discussion Analyzer

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College: BPIT

Course: Major Project (ES-452)

AI-Powered | Team Collaboration | Smart Analysis

Problem We Are Solving

Teams communicate using chats

(WhatsApp, Teams, Slack)

Important decisions get lost

in long conversations

No structured output

from discussions

Users waste time reading

entire chat histories

Team Chat Overload

100+ messages in one day

? Where's the decision?

Lost in conversation thread

Wasted Time

30 mins to find 1 decision

Why This Is Important

- Poor productivity in teams
- Miscommunication and missed tasks
- No accountability for decisions
- Increasing use of online collaboration tools

> *"Teams lose valuable time and insights in unstructured communication"*

Multiple Chats
Confusion

Unclear Goals
Misalignment

Wasted Time
Low Efficiency

Missed Tasks
Deadlines Lost

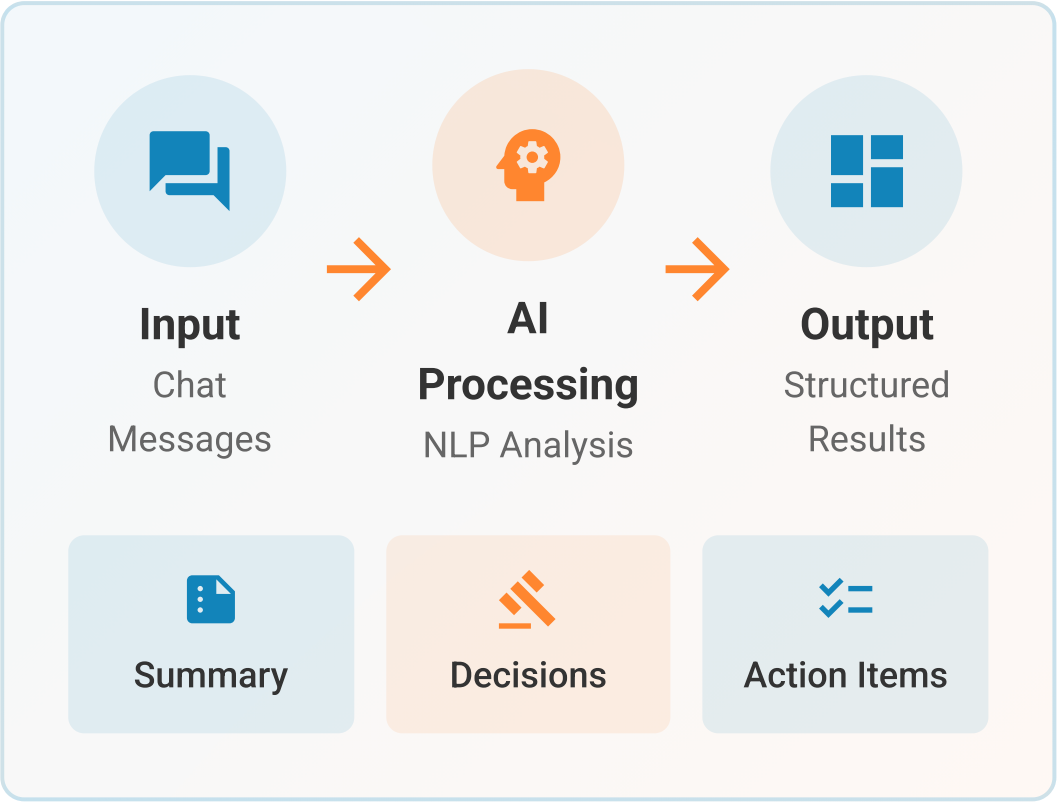
Our Solution

Turning unstructured conversations into structured insights

AI-based system to analyze discussions

Converts chats into structured outputs

- Discussion Summary
- Key Decisions
- Action Items



What We Built

- Web-based collaborative platform
- Users create teams and chat
- AI analyzes discussion
- Displays structured results

Complete solution for team communication and AI-powered analysis

System Flow

1. User Login
2. Create / Join Team
3. Send Messages
4. AI Analysis
5. View Results

System Architecture

Frontend
User Interface

Backend
REST API

NLP Engine
Text Processing

Database
Data Storage

Modular architecture for scalability and maintainability

Architecture Diagram

User
Interacts with system

Frontend UI
React / HTML Interface

Backend API
FastAPI Server

NLP Engine
spaCy / Transformers

Database
SQLite / PostgreSQL

User Data
Messages
Analysis Results

User Data → Messages → Analysis Results

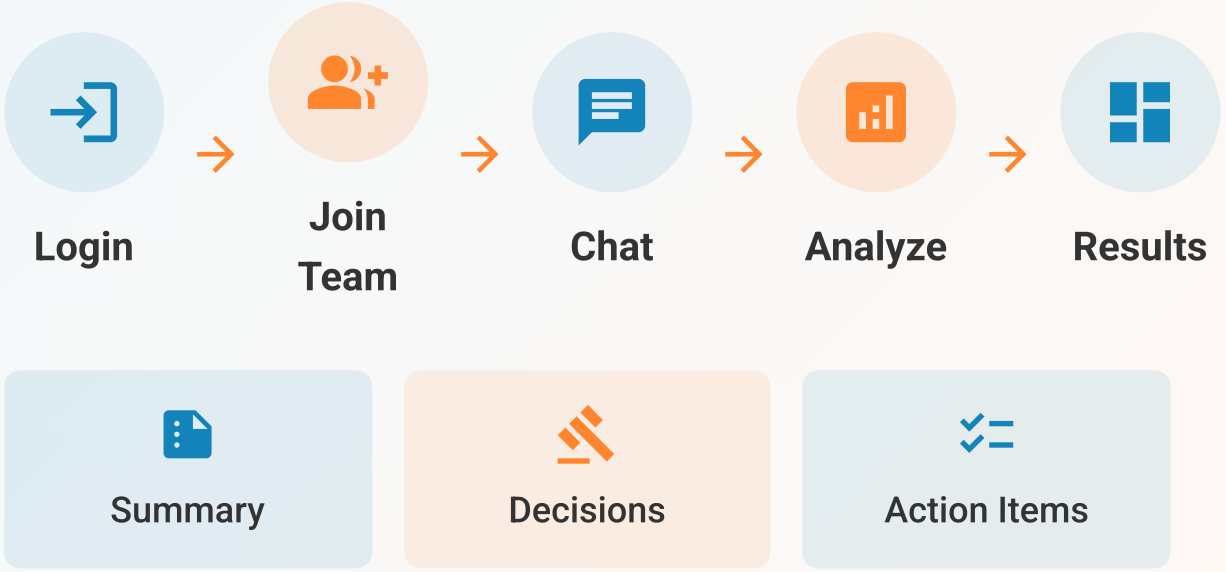
How It Works

Simple 5-Step Process

From team collaboration to actionable insights

1. User joins team
2. Sends messages
3. Clicks "Analyze"
4. AI processes discussion
5. Results displayed

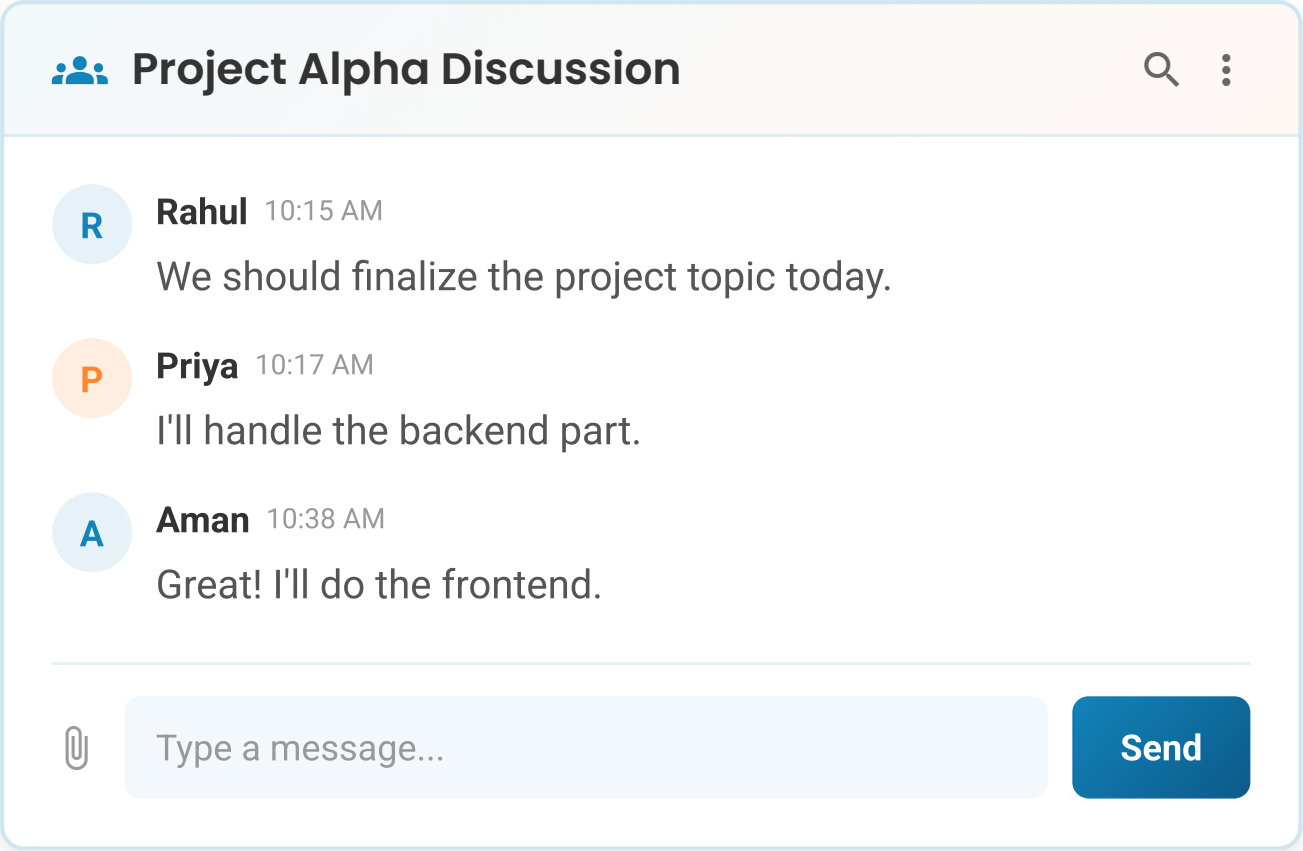
System Workflow



User Interface – Discussion

- Real-time team communication
- Clean chat interface
- Easy to use

Intuitive interface for seamless team collaboration and discussion



AI Generated Output

System Generates:

Summary
Concise overview of discussion

Decisions
Key decisions identified

Action Items
Tasks and assignments extracted

AI transforms chat logs into structured, actionable outputs

AI Analysis Results

Discussion Summary

Team decided to focus on project theme and assigned roles. Backend development assigned to Rahul, documentation to Priya, and initial UI to Aman.

Key Decisions

- Project topic finalized today
- Rahul assigned to backend development

Action Items

- Priya: Prepare documentation
- Aman: Create initial UI design
- Deadline: Tomorrow

Key Features

Automated Summarization

AI generates concise summaries of lengthy discussions

Decision Extraction

Identifies key decisions made during team conversations

Task Identification

Automatically extracts action items and assignments

Easy-to-Use Interface

Intuitive design for seamless team collaboration

Technologies Used

Python
Backend & AI Processing

FastAPI
REST API Framework

NLP Engine
spaCy / Transformers

Frontend
React / HTML

Database
SQLite / PostgreSQL

Architecture
Modular & Scalable

Results

Successfully Extracts Summaries

AI generates accurate and concise discussion summaries

Identifies Decisions & Tasks

Automatically extracts key decisions and action items

Reduces Manual Effort

Saves time by automating discussion analysis

Improves Team Productivity

Enhances collaboration efficiency and output quality

Limitations

Clear Input Required

Works best with well-structured and clear conversation input

Limited Dataset Testing

Testing conducted on a limited set of discussion scenarios

Basic NLP Model

Current implementation uses foundational NLP techniques

Future Work

Real-Time Analysis

Process discussions instantly as messages are sent

Slack/Teams Integration

Seamlessly integrate with popular collaboration platforms

Advanced AI Models

Implement sophisticated transformer-based architectures

Scalable Deployment

Cloud-based architecture for enterprise-scale usage

Conclusion

Solves Real-World Problem

Addresses team collaboration challenges effectively

Demonstrates AI Usefulness

Showcases practical AI applications in business

Scalable & Practical

Ready for enterprise deployment and expansion

> **This project transforms conversations into actionable insights**

Thank You

Questions?